L6: Separating Classes with Hyperplanes and Support Vector Machines

COMP3670/6670
Linear Algebra in Machine Learning
Yun Kuen Cheung, School of Computing, ANU

For some ML tasks, the required prediction is a category, not a number. An email filter, for example, must classify a new email as spam or not spam using features such as word frequencies. Given labelled training emails, how should we choose the weights of such a classifier?

We will first describe a classifier by a hyperplane that separates two classes. The dot products and orthogonal projections from Lecture 5 will let us measure how far each datapoint lies from this decision boundary. We will then use these distances to choose a separator, leading to a *support vector machine* (SVM). Finally, we will allow violations of the separation requirements so that the model can handle overlapping classes and imperfect labels.

Affine scores classify datapoints

For binary classification, the training data consist of n datapoints $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^d$ with labels $y_1, \ldots, y_n \in \{-1, +1\}$. The two numbers encode the two classes; their magnitudes do not represent measured quantities. We seek weights that give positive scores to one class and negative scores to the other. A constant offset allows the boundary between the classes to move away from the origin.

Definition 1. For a weight vector $\mathbf{w} \in \mathbb{R}^d$ and a bias $b \in \mathbb{R}$, an affine score is

$$f_{\mathbf{w},b}(\mathbf{x}) \;=\; \langle \mathbf{w}, \mathbf{x} \rangle + b.$$

The corresponding binary classifier predicts

$$\widehat{y}(\mathbf{x}) \;=\; \begin{cases} +1, & f_{\mathbf{w},b}(\mathbf{x}) \geq 0, \\ -1, & f_{\mathbf{w},b}(\mathbf{x}) < 0. \end{cases}$$

When $\mathbf{w} \neq \mathbf{0}$, the set

$$\mathcal{H} \;=\; \{\mathbf{x} \in \mathbb{R}^d : \langle \mathbf{w}, \mathbf{x} \rangle + b = 0\}$$

is an affine hyperplane, and $\mathbf{w}$ is a normal vector to it. In the context of binary classification, we often call this hyperplane the decision boundary.

An affine hyperplane is a line in $\mathbb{R}^2$ and a plane in $\mathbb{R}^3$. The word "normal" means perpendicular: if $\mathbf{u}, \mathbf{v} \in \mathcal{H}$, then $\langle \mathbf{w}, \mathbf{u} - \mathbf{v} \rangle = \langle \mathbf{w}, \mathbf{u} \rangle - \langle \mathbf{w}, \mathbf{v} \rangle = -b - (-b) = 0$. Thus $\mathbf{w}$ is orthogonal to every displacement along the hyperplane. When $b = 0$, the hyperplane passes through the origin and is a subspace. When $b \neq 0$, it does not contain the origin and is not a subspace.

Example 2. Consider the weights $\mathbf{w} = (3, 4)^\mathsf{T}$ and bias $b = -30$. The score and decision boundary are

$$f_{\mathbf{w},b}(\mathbf{x}) \;=\; 3x_1 + 4x_2 - 30, \qquad \mathcal{H} \;=\; \{\mathbf{x} \in \mathbb{R}^2 : 3x_1 + 4x_2 = 30\}.$$

For three datapoints, we obtain

$\mathbf{x}$	$f_{\mathbf{w},b}(\mathbf{x})$	location
$(6,8)^\mathsf{T}$	$3\cdot 6+4\cdot 8-30=20$	positive side
$(3,4)^\mathsf{T}$	$3\cdot 3+4\cdot 4-30=-5$	negative side
$(2,6)^\mathsf{T}$	$3\cdot 2+4\cdot 6-30=0$	on the boundary

The first two datapoints receive predictions $+1$ and -1, respectively. The final datapoint receives $+1$ under our tie convention, but a tiny displacement to the negative side would change its prediction.

Distance gives scores a geometric meaning

If several hyperplanes classify every training datapoint correctly, we still need a reason to prefer one. The prediction for a datapoint close to the boundary can change under a small perturbation of its features. This motivates measuring the distance to the boundary. The raw score is not itself a distance: multiplying both $\mathbf{w}$ and b by a positive constant rescales the scores without moving the hyperplane.

Proposition 3. Let $\mathcal{H}=\{\mathbf{z}:\langle \mathbf{w},\mathbf{z}\rangle+b=0\}$ with $\mathbf{w}\neq\mathbf{0}$. For every $\mathbf{x}\in\mathbb{R}^d$, its unique closest point on $\mathcal{H}$ is

$$\mathbf{p} \;=\; \mathbf{x}-\frac{\langle \mathbf{w},\mathbf{x}\rangle+b}{\|\mathbf{w}\|_2^2}\mathbf{w}.$$

The distance from $\mathbf{x}$ to $\mathcal{H}$ and its signed distance, taken as positive on the side towards which $\mathbf{w}$ points, are respectively

$$d(\mathbf{x},\mathcal{H}) \;=\; \frac{|\langle \mathbf{w},\mathbf{x}\rangle+b|}{\|\mathbf{w}\|_2}, \qquad s(\mathbf{x},\mathcal{H}) \;=\; \frac{\langle \mathbf{w},\mathbf{x}\rangle+b}{\|\mathbf{w}\|_2}.$$

Proof: Substitution shows that $\mathbf{p}$ lies on the hyperplane:

$$\langle \mathbf{w},\mathbf{p}\rangle+b \;=\; \langle \mathbf{w},\mathbf{x}\rangle+b-\frac{\langle \mathbf{w},\mathbf{x}\rangle+b}{\|\mathbf{w}\|_2^2}\langle \mathbf{w},\mathbf{w}\rangle \;=\; 0.$$

Moreover, $\mathbf{x}-\mathbf{p}$ is a multiple of $\mathbf{w}$. For every $\mathbf{z}\in\mathcal{H}$, the displacement $\mathbf{p}-\mathbf{z}$ is orthogonal to $\mathbf{w}$, so $\langle \mathbf{x}-\mathbf{p},\mathbf{p}-\mathbf{z}\rangle=0$. Expanding the squared norm therefore gives

$$\|\mathbf{x}-\mathbf{z}\|_2^2 \;=\; \|(\mathbf{x}-\mathbf{p})+(\mathbf{p}-\mathbf{z})\|_2^2 \;=\; \|\mathbf{x}-\mathbf{p}\|_2^2+\|\mathbf{p}-\mathbf{z}\|_2^2 \;\geq\; \|\mathbf{x}-\mathbf{p}\|_2^2,$$

with equality exactly when $\mathbf{z}=\mathbf{p}$. Hence $\mathbf{p}$ is the unique closest point. Finally,

$$\|\mathbf{x}-\mathbf{p}\|_2 \;=\; \frac{|\langle \mathbf{w},\mathbf{x}\rangle+b|}{\|\mathbf{w}\|_2^2}\|\mathbf{w}\|_2 \;=\; \frac{|\langle \mathbf{w},\mathbf{x}\rangle+b|}{\|\mathbf{w}\|_2}.$$

□

Example 4. For $\mathbf{w}=(3,4)^\mathsf{T}$, $b=-30$ and $\mathbf{x}=(6,8)^\mathsf{T}$ from Example 2, we have $\|\mathbf{w}\|_2=5$ and score 20. The signed distance is $20/5=4$, and the closest point on the boundary is

$$\mathbf{p} \;=\; \begin{pmatrix}6\\8\end{pmatrix}-\frac{20}{25}\begin{pmatrix}3\\4\end{pmatrix} \;=\; \begin{pmatrix}18/5\\24/5\end{pmatrix}.$$

Indeed, $3(18/5)+4(24/5)=30$. If we replace the weights and bias by $\mathbf{w}'=(21,28)^\mathsf{T}$ and $b'=-210$, the score becomes 140, but the signed distance remains $140/35=4$. The new parameters describe exactly the same classifier.

For a labelled datapoint, distance should count as positive only when the datapoint lies on the correct side. Multiplying the score by the label makes the two classes obey the same condition.

Definition 5. For a labelled datapoint $(\mathbf{x}_i, y_i)$ and $\mathbf{w} \neq \mathbf{0}$, its functional margin and geometric margin are

$$m_i \;=\; y_i\big(\langle \mathbf{w}, \mathbf{x}_i\rangle + b\big), \qquad \gamma_i \;=\; \frac{m_i}{\|\mathbf{w}\|_2}.$$

The datapoint lies strictly on the correct side exactly when $m_i > 0$, and strictly on the wrong side exactly when $m_i < 0$. Its geometric margin is its distance to the boundary with these signs.

The dataset is linearly separable if some $\mathbf{w}, b$ satisfy $m_i > 0$ for every i. The margin of a separating classifier is

$$\gamma \;=\; \min_{i=1,\dots,n} \gamma_i.$$

The geometric margin has a direct robustness interpretation. If $\|\boldsymbol{\delta}\|_2 < \gamma_i$ for a correctly classified training datapoint, Cauchy–Schwarz gives

$$y_i f_{\mathbf{w},b}(\mathbf{x}_i + \boldsymbol{\delta}) \;=\; m_i + y_i \langle \mathbf{w}, \boldsymbol{\delta}\rangle \;\geq\; m_i - \|\mathbf{w}\|_2\|\boldsymbol{\delta}\|_2 \;>\; 0.$$

Thus its prediction cannot change under such a perturbation. A larger training margin does not, by itself, guarantee correct predictions on unseen data, but it gives a geometric reason to prefer a separator that does not pass too close to the training datapoints.

A support vector machine maximises the margin

We now seek a separator whose closest training datapoint is as far from the boundary as possible. Assume that both classes occur and that the dataset is linearly separable. To set a common reference scale for the scores, we will require every functional margin to be at least 1. These constraints guarantee a geometric margin of at least $1/\|\mathbf{w}\|_2$. We will show that minimising the weight norm under these constraints maximises the margin.

Definition 6. The hard-margin support vector machine chooses $\mathbf{w} \in \mathbb{R}^d$ and $b \in \mathbb{R}$ to solve

$$\begin{aligned} \text{minimise} \quad & \frac{1}{2}\|\mathbf{w}\|_2^2 \\ \text{subject to} \quad & y_i\big(\langle \mathbf{w}, \mathbf{x}_i\rangle + b\big) \;\geq\; 1 \qquad (i = 1, \dots, n). \end{aligned}$$

At an optimal solution, training datapoints satisfying $y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b) = 1$ are called support vectors. They lie on the two margin boundaries $\langle \mathbf{w}, \mathbf{x}\rangle + b = 1$ and $\langle \mathbf{w}, \mathbf{x}\rangle + b = -1$.

Proposition 7. For a linearly separable dataset containing both classes, the hard-margin SVM is equivalent to maximising the geometric margin. At an optimal solution,

$$\min_{i=1,\dots,n} y_i\big(\langle \mathbf{w}, \mathbf{x}_i\rangle + b\big) \;=\; 1, \qquad \gamma \;=\; \frac{1}{\|\mathbf{w}\|_2}.$$

The distance between the two margin boundaries is $2/\|\mathbf{w}\|_2$.

Proof: Start with any strict separator and let $t = \min_i y_i(\langle \mathbf{w}, \mathbf{x}_i \rangle + b) > 0$. Rescale both parameters by $1/t$:

$$\widetilde{\mathbf{w}} \;=\; \frac{\mathbf{w}}{t}, \qquad \widetilde{b} \;=\; \frac{b}{t}.$$

The decision boundary and geometric margins do not change, while the smallest functional margin becomes 1. This gives a feasible solution of the hard-margin constraints with

$$\gamma \;=\; \frac{t}{\|\mathbf{w}\|_2} \;=\; \frac{1}{\|\widetilde{\mathbf{w}}\|_2}.$$

Consequently, among separators with smallest functional margin 1, maximising γ is equivalent to minimising the norm, or half its square.

It remains to check that the constraints do not select a different scale at an optimum. Both classes occur, so a feasible solution cannot have $\mathbf{w} = \mathbf{0}$: it would require both $b \geq 1$ and $b \leq -1$. If a feasible solution had smallest functional margin $t > 1$, dividing both $\mathbf{w}$ and b by t would give another feasible solution and would strictly decrease $\frac{1}{2}\|\mathbf{w}\|_2^2$. Hence an optimum has $t = 1$.

Each margin boundary is at distance $1/\|\mathbf{w}\|_2$ from the decision boundary, on opposite sides, by Proposition 3. Their separation is therefore $2/\|\mathbf{w}\|_2$. □

The constraints keep every training datapoint outside the open region between the two margin boundaries. The support vectors touch these boundaries and prevent the margin from being widened further without moving or rotating the separator.

Example 8. Consider the four labelled datapoints

i	$\mathbf{x}_i$	y_i
1	$(9, 12)^\mathsf{T}$	$+1$
2	$(16, 13)^\mathsf{T}$	$+1$
3	$(3, 4)^\mathsf{T}$	-1
4	$(4, -3)^\mathsf{T}$	-1

Take

$$\mathbf{w} \;=\; \frac{1}{25}\begin{pmatrix}3\\4\end{pmatrix}, \qquad b \;=\; -2.$$

The scores are $1, 2, -1, -2$, respectively, so the functional margins are $1, 2, 1, 2$. The constraints are satisfied, and $\mathbf{x}_1, \mathbf{x}_3$ are support vectors. Since $\|\mathbf{w}\|_2 = 1/5$, the geometric margin is 5. The decision boundary and margin boundaries are

$$3x_1 + 4x_2 \;=\; 50, \qquad 3x_1 + 4x_2 \;=\; 75, \qquad 3x_1 + 4x_2 \;=\; 25.$$

We can prove that this choice is optimal without running an optimisation algorithm. Every feasible pair $\mathbf{w}', b'$ must satisfy the constraints for $\mathbf{x}_1$ and $\mathbf{x}_3$:

$$\langle \mathbf{w}', \mathbf{x}_1 \rangle + b' \;\geq\; 1, \qquad \langle \mathbf{w}', \mathbf{x}_3 \rangle + b' \;\leq\; -1.$$

Subtracting and using Cauchy–Schwarz gives

$$2 \;\leq\; \langle \mathbf{w}', \mathbf{x}_1 - \mathbf{x}_3 \rangle \;\leq\; \|\mathbf{w}'\|_2 \|\mathbf{x}_1 - \mathbf{x}_3\|_2 \;=\; 10\|\mathbf{w}'\|_2.$$

Thus every feasible weight vector has norm at least $1/5$. Our choice attains this lower bound, so it solves the hard-margin SVM.

Soft margins allow violations

Real training data may contain overlapping classes, measurement noise or incorrect labels. Then no hyperplane may satisfy all the hard-margin constraints. Even when strict separation is possible, one unusual datapoint may force a very narrow margin. A *soft-margin SVM* allows violations but charges for them, balancing the preference for a wide margin against the fit to the labelled data.

Definition 9. Fix a parameter $C > 0$. The soft-margin support vector machine solves

$$\begin{aligned} \text{minimise} \quad & \frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{i=1}^{n} \xi_i \\ \text{subject to} \quad & y_i\big(\langle \mathbf{w}, \mathbf{x}_i\rangle + b\big) \;\geq\; 1 - \xi_i \qquad (i = 1, \ldots, n), \\ & \xi_i \;\geq\; 0 \qquad (i = 1, \ldots, n). \end{aligned}$$

$\xi_1, \ldots, \xi_n$ are slack variables; each permits a violation of one datapoint's unit functional-margin requirement. The hinge loss of a labelled score with functional margin m is $\ell(m) = \max\{0, 1 - m\}$.

Proposition 10. For fixed $\mathbf{w}, b$, the best choice of slack variables is

$$\xi_i \;=\; \max\{0, 1 - y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b)\}.$$

Thus the soft-margin SVM is equivalently the unconstrained minimisation of

$$\frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{i=1}^{n} \ell\big(y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b)\big).$$

Proof: For each i, the two constraints require both $\xi_i \geq 0$ and $\xi_i \geq 1 - y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b)$. Thus, the smallest permitted value of ξ_i is $\max\{0, 1 - y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b)\}$. Since $C > 0$, substituting these smallest values gives the hinge-loss objective. □

The following geometric interpretation refers to these smallest slack values and assumes $\mathbf{w} \neq \mathbf{0}$. A datapoint with $\xi_i = 0$ meets the unit margin requirement. If $0 < \xi_i < 1$, it is strictly on the correct side but inside the margin. If $\xi_i = 1$, it lies on the decision boundary. If $\xi_i > 1$, it lies strictly on the wrong side. In particular, *hinge loss penalises some correctly classified datapoints as well as misclassified ones.*

Example 11. Keep $\mathbf{w} = (3, 4)^{\mathsf{T}}/25$ and $b = -2$. For the following labelled datapoints, the resulting scores, functional margins and smallest slack values are:

$\mathbf{x}_i$	y_i	$f_{\mathbf{w},b}(\mathbf{x}_i)$	$m_i = y_i f_{\mathbf{w},b}(\mathbf{x}_i)$	$\xi_i = \ell(m_i)$
$(12, 16)^{\mathsf{T}}$	$+1$	2	2	0
$(8, 9)^{\mathsf{T}}$	$+1$	$2/5$	$2/5$	$3/5$
$(6, 8)^{\mathsf{T}}$	-1	0	0	1
$(0, 0)^{\mathsf{T}}$	$+1$	-2	-2	3

The second datapoint is correctly classified but incurs a positive loss because it is too close to the boundary. The fourth is misclassified and incurs loss 3. These are the losses of this particular classifier; we have not claimed that it is optimal for the new dataset.

A larger C places more weight on reducing margin violations relative to the norm penalty, while a smaller C places more weight on keeping the weights small. Unlike the hard-margin problem, the soft-margin problem permits $\mathbf{w} = \mathbf{0}$, which gives a constant score rather than a genuine hyperplane.

The geometry depends on the feature representation: changing a feature's units changes Euclidean distances and can change the learned separator. Feature scaling and the choice of C should therefore be considered together.

Computing the SVM parameters

The optimisation problem specifies which parameters we want, but it does not yet tell us how to find them. For a large dataset, we cannot simply guess a separator as we did in Example 8. Instead, numerical methods repeatedly adjust the parameters to approach the minimum objective value.

Both constrained SVM formulations have a quadratic objective and linear inequality constraints, so they can be handled by standard *quadratic-programming (QP) solvers*. Given the training data and, and C for a soft-margin SVM, such software computes the model parameters. We will not study QP algorithms in detail here. Such QP problems belong to *convex optimization*: their objectives are convex and their constraints are linear. Consequently, every local minimum is also a global minimum: there are no inferior local minima in which a solver could become trapped. This does not mean that arbitrary parameter updates will converge; a suitable numerical method is still needed. You will learn some basics of convex optimization later in this course. A fuller treatment is beyond the scope of this course.

For the soft-margin SVM, we can also work directly with the hinge-loss objective in Proposition 10. Its two terms have competing effects. The squared-norm term favours smaller weights, while datapoints whose functional margins are below 1 call for corrections that move their scores towards their labelled sides. The following update illustrates how an iterative method can combine these effects.

Example 12. At the current parameters $\mathbf{w}, b$, let

$$\mathcal{I} \;=\; \{i : y_i(\langle \mathbf{w}, \mathbf{x}_i \rangle + b) < 1\}$$

be the indices of datapoints violating the unit margin requirement. One possible update, with a small step size $0 < \eta < 1$, is

$$\mathbf{w}_{\text{new}} \;=\; (1-\eta)\mathbf{w} + \eta C \sum_{i \in \mathcal{I}} y_i \mathbf{x}_i,$$

$$b_{\text{new}} \;=\; b + \eta C \sum_{i \in \mathcal{I}} y_i.$$

Both expressions use the same set $\mathcal{I}$, computed before either parameter is changed. The factor $1 - \eta$ shrinks the weights. Each violating datapoint contributes a correction in the direction $y_i \mathbf{x}_i$ to the weights and in the direction y_i to the bias. Considered on its own, this correction increases that datapoint's functional margin by $\eta C(\|\mathbf{x}_i\|_2^2 + 1)$. Corrections from other datapoints and the weight shrinkage may counteract this increase, so the update balances the requirements rather than treating each datapoint in isolation.

This is an example of a *subgradient method*, which accommodates the corner in the hinge-loss function at margin 1. The displayed update makes no loss-based correction for datapoints exactly at that corner, which is one valid choice. Software recomputes the margins and repeats the updates, using suitable step-size rules and numerical stopping criteria. Individual steps need not decrease the objective, and convergence is not guaranteed for an arbitrary fixed step size. The precise convergence conditions are beyond the scope of this course.